

Omar Sherif Elghamry

Backend & AI Engineer

Giza, Egypt omarsherifelghamry@gmail.com linkedin.com/in/omar-elghamry github.com/omaaarsh

Backend Software Engineer focused on Node.js/NestJS and Python/FastAPI, with hands-on production experience across e-commerce, payment systems, event-driven architecture, and AI-powered backend services. I own features end-to-end — from architecture and implementation through testing, deployment, and production support. Currently deepening distributed systems, system design, observability, and performance.

EXPERIENCE

Software Engineer · Techkhana 2026 — Present · Full-time

Backend engineering on real production systems for a Saudi-market super-app (Barns).

- Refactored payment architecture around a provider abstraction (factory) and integrated Tap Payments alongside PayFort — charge, save-card, and authorize flows.
- Designed an e-commerce checkout that snapshots catalog data, with backend-to-backend order coordination between services.
- Worked with Apache Kafka and Kafka Connect (CDC) for change-data-capture synchronization.
- Built admin RBAC (roles, permissions, guards) and a centralized image-processing service (magic-byte validation, WebP, SHA-256 dedup over S3).
- Integrated an AI feature ("Fahem") using AWS Rekognition; ran services on Kubernetes (EKS) with Docker and GitHub Actions.

Backend & AI — Graduation Project · StyleHub 2025 — 2026 · Grade A*

AI-driven fashion platform (social media, e-commerce, recommendations).

- Owned the NestJS Social Media Service — feeds, posts, follow graph, chat, notifications; RabbitMQ events, Socket.IO real-time, JWT + RBAC; 12 test suites / 161 tests passing.
- Built the Python FastAPI AI service — LangChain, LangGraph, RAG, and multi-agent orchestration over OpenAI and AWS Bedrock (Claude).

TECHNICAL SKILLS

Backend: Node.js, NestJS, TypeScript, Python, FastAPI	Data & Caching: PostgreSQL, TypeORM, Redis
Messaging & Real-time: Apache Kafka, RabbitMQ, Socket.IO	Infra & Cloud: Docker, Kubernetes (EKS), AWS, GitHub Actions
AI Integration: LangChain, LangGraph, RAG, AWS Bedrock, Claude	Testing: Jest, Testcontainers, Postman

SELECTED PROJECTS

- Social Media Service** — Production NestJS microservice: 59 endpoints, 14 modules, RabbitMQ, Socket.IO, 161 tests. NestJS · PostgreSQL · RabbitMQ
- StyleHub Chat Frontend** — Real-time chat & social UI (TanStack Query, Zustand, Socket.IO). React · TypeScript
- LINCS Gene Query Tool** — Streamlit app for gene→drug reverse-search with Plotly charts and CSV export. Python · Streamlit
- Deep Neural Network** — Binary classifier built from scratch in Python/NumPy (forward + backprop). Python · NumPy

EDUCATION

B.Sc. Computer Science · Misr University for Science and Technology (MUST) Graduating June 2026
CGPA 3.8 / 4.0 · Graduation project graded A*